

Lecture 4 Parametric Curves

$$\vec{r}(t) = f(t)\vec{i} + g(t)\vec{j} + h(t)\vec{k}$$

$$\vec{r}'(t) = f'(t)\vec{i} + g'(t)\vec{j} + h'(t)\vec{k}$$

Properties. $f(t)$, $\vec{u}(t)$, $\vec{v}(t)$

$$1) \frac{d}{dt} (f(t) \vec{u}(t)) = f'(t) \vec{u}(t) + f(t) \vec{u}'(t)$$

$$2) \frac{d}{dt} (\vec{u}(t) \cdot \vec{v}(t)) = \vec{u}'(t) \cdot \vec{v}(t) + \vec{u}(t) \cdot \vec{v}'(t)$$

$$3) \frac{d}{dt} (\vec{u}(t) \times \vec{v}(t)) = \vec{u}'(t) \times \vec{v}(t) + \vec{u}(t) \times \vec{v}'(t)$$

$$4) \frac{d}{dt} (\vec{u}(f(t))) = \vec{u}'(f(t)) f'(t) \quad \text{chain rule}$$

Pf of 2). $\vec{u}(t) = \langle u_1(t), u_2(t), u_3(t) \rangle$

$$\vec{v}(t) = \langle v_1(t), v_2(t), v_3(t) \rangle$$

$$\frac{d}{dt} (\vec{u}(t) \cdot \vec{v}(t)) = \frac{d}{dt} (u_1(t)v_1(t) + u_2(t)v_2(t) + u_3(t)v_3(t))$$

$$= \sum_{i=1}^3 (u_i'(t)v_i(t) + u_i(t)v_i'(t))$$

$$= \vec{u}'(t) \cdot \vec{v}(t) + \vec{u}(t) \cdot \vec{v}'(t)$$

Example Let $\vec{r}(t)$ represent a particle travelling at uniform speed C .

Show that its velocity and acceleration vectors are orthogonal.

Pf. $|\vec{r}'(t)| = C \Rightarrow \vec{r}'(t) \perp \vec{r}''(t)$

$$\Downarrow$$

$$|\vec{r}'(t)|^2 = C^2 \Rightarrow \vec{r}'(t) \cdot \vec{r}'(t) = C^2 \Rightarrow \frac{d}{dt} (\vec{r}'(t) \cdot \vec{r}'(t)) = 0 \Rightarrow 2\vec{r}'(t) \cdot \vec{r}''(t) = 0$$



Integrals / Arc Lengths of Curves.

$$\text{distance} = \text{speed} \times \text{time} \Rightarrow \text{distance} = \int \text{speed} \times d(\text{time})$$



$$\text{area} = \text{height} \times \text{width} \Rightarrow \text{area} = \int \text{height} \times d(\text{width})$$

Theorem. Given $\vec{r}(t)$. The arc length of the curve from the point at $t=a$ to the point at $t=b$ is given by:

$$\text{arc length} = \int_a^b |\vec{r}'(t)| dt.$$

Example. Find the arc length of the curve $\vec{r}(t) = \frac{1}{2}t^2 \vec{i} + \frac{2\sqrt{t}}{3} \vec{j} + t \vec{k}$ from $(0,0,0)$ to $(2, \frac{8}{3}, 2)$.

Sol. $\vec{r}(0) = \langle 0, 0, 0 \rangle$, $\vec{r}(2) = \langle 2, \frac{8}{3}, 2 \rangle$

$$\vec{r}'(t) = \langle t, \sqrt{t}, 1 \rangle$$

$$|\vec{r}'(t)| = \sqrt{t^2 + 2t + 1} = \sqrt{(t+1)^2} = t+1$$

$$\int_0^2 |\vec{r}'(t)| dt = \int_0^2 (t+1) dt = \left. \frac{t^2}{2} + t \right|_0^2 = 4$$

$$\vec{r}_1(t) = (\cos t) \vec{i} + (\sin t) \vec{j} + t \vec{k} \quad 0 \leq t \leq 2\pi$$

$$\vec{r}_2(t) = (\cos 2t) \vec{i} + (\sin 2t) \vec{j} + 2t \vec{k} \quad 0 \leq t \leq \pi$$

Arc-length parametrization: we say the curve $\vec{r}(s)$ is parametrized by arc-length if $|\vec{r}'(s)| = 1$.

Procedures to find the arc-length parametrization.

Need to show $|\hat{\vec{r}}(s)| = 1$

1° Given a curve $\vec{r}(t): [a,b] \rightarrow \mathbb{R}^3$, compute the following integral $\vec{\tilde{r}}(s) = \vec{r}(t(s))$

$$s = \int_a^t |\vec{r}'(z)| dz = f(t) \quad t = f^{-1}(s)$$

$$\begin{aligned} |\vec{\tilde{r}}(s)| &= \left| \frac{d\vec{r}(t(s))}{ds} \right| \\ &= \left| \frac{d\vec{r}}{dt} \frac{dt}{ds} \right| \end{aligned}$$

2° Express t in terms s , i.e. $t = t(s) = f^{-1}(s)$

3° Replace all t 's by $t(s)$ in the curve $\vec{r}(t)$

$$= |\vec{r}'(t)| \left| \frac{dt}{ds} \right|$$

By Fundamental Theorem of Calculus.

$$\frac{ds}{dt} = |\vec{r}'(t)|$$

$$\frac{dt}{ds} = \frac{1}{|\vec{r}'(t)|}$$

$$\Rightarrow \left| \frac{d}{ds} \right| = 1$$

Example. Find the arc-length parametrization of the curve

$$\vec{r}(t) = (\cos t) \vec{i} + (\sin t) \vec{j} + t \vec{k} \quad t \in [0, 2\pi]$$

Sol. $|\vec{r}'(t)| = \sqrt{2}$

$$s(t) = \int_0^t |\vec{r}'(z)| dz = \int_0^t \sqrt{2} dz = \sqrt{2} t$$

$$\Rightarrow t = \frac{s}{\sqrt{2}}$$

$$\vec{r}(s) = \left(\cos \frac{s}{\sqrt{2}} \right) \vec{i} + \left(\sin \frac{s}{\sqrt{2}} \right) \vec{j} + \frac{s}{\sqrt{2}} \vec{k}$$

Curvature.



Def. Given a curve γ in $\mathbb{R}^1$ or $\mathbb{R}^3$ which can be arc-length parametrized by $\vec{r}(s)$, then its curvature is given by $\kappa(s) = |\vec{r}''(s)|$

The magnitude of the rate at which the direction changes with respect to arc length.

Example. Find the curvature of the circle of radius R centered at the origin $(0,0)$ in $\mathbb{R}^2$.

Sol. $\vec{r}(t) = (R \cos t, R \sin t) \Rightarrow |\vec{r}'(t)| = R$

$$s(t) = \int_0^t |\vec{r}'(z)| dz = \int_0^t R dz = Rt \Rightarrow t = \frac{s}{R}$$

$$\vec{r}(s) = \left(R \cos \frac{s}{R}, R \sin \frac{s}{R} \right)$$

$$\vec{r}'(s) = \left(-\sin \frac{s}{R}, \cos \frac{s}{R} \right)$$

$$\hat{r}''(s) = \left(-\frac{1}{R} \cos \frac{s}{R}, -\frac{1}{R} \sin \frac{s}{R}\right)$$

$$k(s) = |\hat{r}''(s)| = \frac{1}{R}$$

Theorem (Curvature Formula for General Curves)

Let $\vec{r}(t)$ denote a parametric curve. If $\vec{v} = \vec{r}'(t)$ is the velocity and

$\vec{a} = \vec{v}'$ is the acceleration, then the curvature is

$$k(t) = \frac{|\vec{v} \times \vec{a}|}{|\vec{v}|^3}$$

Theorem 12.11 from Calculus for Scientists and Engineers.

Q: How can this general formula be reduced to the previous one?

$$|\vec{v}| = |\vec{r}'| = 1$$

If $|\vec{v}| = C$, then $\vec{v} \perp \vec{a}$

$$|\vec{v} \times \vec{a}| = |\vec{v}| |\vec{a}| \sin \theta = |\vec{a}| \sin \theta = |\vec{a}| = |\vec{r}''|$$

θ is the angle between $\vec{v}$ and $\vec{a}$

Eg. $\vec{r}(t) = R \cos t \vec{i} + R \sin t \vec{j} + 0 \vec{k}$

$$\vec{v}(t) = \vec{r}'(t) = -R \sin t \vec{i} + R \cos t \vec{j}$$

$$\vec{a}(t) = \vec{v}'(t) = -R \cos t \vec{i} - R \sin t \vec{j}$$

$$\vec{v} \times \vec{a} = R^2 \sin^2 t \vec{k} + R^2 \cos^2 t \vec{k} = R^2 \vec{k}$$

$$|\vec{v} \times \vec{a}| = R^2 \quad |\vec{v}| = R$$

$$\Rightarrow k(t) = \frac{|\vec{v} \times \vec{a}|}{|\vec{v}|^3} = \frac{R^2}{R^3} = \frac{1}{R}$$